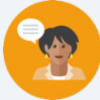


11.9 Solving Problems Using Triangle Relationships

Lesson Introduction

 Benchmark Focus	MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.		 Learning Target	I can solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.		MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.	
 Essential Questions	How can you use what you know about triangle relationships, postulates, and theorems to solve mathematical and real-world problems?			
 Lesson Narrative	In this lesson, students consolidate and apply the knowledge and skills they obtained in the previous eight lessons. They use the triangle relationships, postulates, and theorems learned throughout the unit to solve mathematical and real-world problems.			
 Component Summary	11.9.1 Warm-Up (~5 min.)	Determine which coupon they would rather use (<i>SE p. 225</i>)	11.9.2 Activity (35-40 min.)	Apply triangle relationships, postulates, and theorems to solve problems (<i>SE p. 226</i>)
	11.9.3 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 227</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> - Triangle Inequality Theorem - Triangle Sum Theorem - Triangle Midsegment Theorem		- Copies of 11.9.2 Activity Questions (Optional) Colored Pencils	
	Additional Preparation			
	N/A			

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may use the wrong relationship, postulate, or theorem when solving a problem. - Students may miscalculate when finding the value or a variable.	11.9.2 Activity is a mixed review of the concepts and procedures explored throughout the unit. Students will benefit from referencing relationships, postulates, and theorems on the classroom Word Wall or anchor charts.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Take Turns In 11.9.2 Activity, students apply relationships, postulates, and/or theorems that they have learned in this unit to solve mathematical and real-world problems. Consider having students work with a partner and take turns explaining how to solve the problem. Prompt students to discuss if they agree or disagree with their partners, and why.	MA.K12.MTR.3.1: Mathematical Fluency Students first explored the relationship, postulates, and theorems in the previous lessons in the unit. 11.9.2 Activity provides an additional opportunity for students to practice applying their knowledge and skills to solve mathematical and real-world problems. This additional practice will help students solidify their knowledge, skills, and confidence with the lesson learning targets from the unit.
	Provide a kinesthetic entry point by structuring 11.9.2 Activity as a circuit. Hang each problem with a different problem’s answer at different stations around the room. Students can start anywhere, but ask them to spread out, so no more than two students at each station. Have them work the problems against the walls or give them a book or clipboard to write on as they stand. Then, have students look around the room for the answer. When they find the answer, prompt them to solve the problem at that station. The activity is self-checking, because if students can complete the circuit, then they answered all of the problems correctly.	
Lesson Notes:		

11.9.1 Warm-Up: Would You Rather?

Component Overview: Students determine which coupon they would rather use and justify their decisions.



11.9.1 Warm-Up: Would You Rather?

1. Would you rather use a coupon worth ...

\$20 off your
entire purchase

OR

25% off your
entire purchase?

Answers will vary. It depends on how large the purchase is.

2. Explain your choice.

Answers will vary. Depending on how large the purchase is, I would choose the option that saves me the most money. Any item below \$80 is better to use the \$20 off. Any item above \$80 is better to use the 25% off.



This task should engage all students' interest because it is connected to saving money, a topic all consumers can benefit from understanding. The better coupon depends on the price of the item. Challenge students to determine the price when one coupon becomes more beneficial than the other.

11.9.2 Activity: Triangle Relationships

Component Overview: Students apply triangle relationships, postulates, and theorems to solve mathematical and real-world problems. Encourage students to show all of their work and check their answers.



11.9.2 Activity: Triangle Relationships

1. For each problem, record your work and answer in the table.

Problem Letter	Work	Answer
A	$9x - 15 + x + 11 + 3x + 2 = 180$ $x = 14$ $9(14) - 15 = 111$	111°
B	$5 + 5 = 10$ $10 < 11$	No, because $5 + 5 < 11$, meaning it does not meet the triangle inequality theorem.
C	$18x - 14 + 6x + 24 + 146 = 360$ $x = 8.5$ $18(8.5) - 14 = 139$	139°
D	$\frac{4}{3}x + 7 = \frac{1}{2}(8x + 2)$ $x = 2.25$ $\frac{4}{3}(2.25) + 7 = 10$	10 meters
E	$YN = 30, YW = 22, NW = 26$ $22 + 30 + 26 = 78$	78 units
F	$3x + 7 + 9x + 1 + 5x - 15 = 180$ $x = 11$ $9(11) + 1 = 100$	100°
G	$2(4x - 1) + 8x + 3 + 2\left(\frac{4}{3}x + 2\right)$ $= 84$ $x = 4.5$	4.5



Provide a kinesthetic entry point by structuring this component as a circuit. Hang each problem with a different problem's answer at different stations around the room. Students can start anywhere, but ask them to spread out, so no more than two students at each station. Have them work the problems against the walls or give them a book or clipboard to write on as they stand. Then, have students look around the room for the answer. When they find the answer, prompt them to solve the problem at that station. The activity is self-checking, because if students can complete the circuit, then they answered all of the problems correctly.



Take Turns

Consider having students work with a partner and take turns explaining how to solve the problem. Prompt students to discuss if they agree or disagree with their partners, and why.

11.9.3 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



11.9.3 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.